

Anna Maddux

Personal

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Research Interests

Primary: Algorithmic game theory; mechanism design; multi-agent learning.
Secondary: Algorithmic pricing and collusion; international trade; general equilibrium theory.

Education

- École Polytechnique Fédérale de Lausanne (EPFL) (Lausanne, Switzerland) | PhD in Applied Mathematics** Feb 2022 – May 2026
Supervisor: Maryam Kamgarpour (Sycamore Lab)
- Thesis: “Game-Theoretic Decision Making in Multi-Agent Systems under Constraints and Welfare Objectives”
 - Committee: Galit Ashkenazi-Golan, Nicolò Cesa-Bianchi, Giancarlo Ferrari Trecate (President), Heinrich Nax
- MIT (Cambridge, USA) | Visiting Fellow** Apr 2025 – Jun 2025
Supervisor: Gabriele Farina (Laboratory for Information and Decision Systems)
- Investigated equilibrium selection mechanisms to resolve coordination failures among independent agents.
- ETH Zürich (Zürich, Switzerland) | M.Sc. in Statistics** Sep 2019 – May 2021
Supervisors: Fadoua Balabdaoui, Giuseppe Belgioioso, Florian Dörfler, Nicolò Pagan
- Thesis: “Behaviour Estimation in Dynamic Games”
- University of Freiburg (Freiburg, Germany) | B.Sc. in Mathematics** Oct 2015 – Mar 2019
- Thesis: “Symmetric Random Walks and a Generalization of the Borel-Cantelli Lemma”
- University of Freiburg (Freiburg, Germany) | B.Sc. in Economics** Oct 2013 – Feb 2019
- Thesis: “Hyperparameter Optimization of Neural Networks: Grid Search versus Random Search”

Research and Professional Experience

- Bank for International Settlements (BIS) | Technical Advisor** May 2026 – Aug 2026
Monetary and Economic Department (MED); full-time
- ETH Zürich | Research Assistant** Oct 2021 – Dec 2021
Computational Social Science (COSS) Group; part-time
- ETH Zürich | Research Assistant** May 2021 – Jul 2021
Automatic Control Laboratory (IfA); part-time
- Daimler-Benz, Stuttgart | Data Science Intern** Apr 2019 – Aug 2019
Advanced Analytics and Big Data Group; full-time
- German Bundestag, Berlin | Research Intern** Jan 2019 – Feb 2019
Dr. Christoph Hoffmann, MdB, Economic Cooperation and Development Committee; full-time

Publications and Working Papers

* Equal contribution. + Mentored student.

Peer-Reviewed Journal Publications

Finite-time convergence to an ϵ -efficient Nash equilibrium in potential games
Anna Maddux^{}, Reda Ouhamma^{*}, Hana Catic, and Maryam Kamgarpour.*
IEEE Transactions on Control of Network Systems (TCNS), 2026.

Peer-Reviewed Conference and Workshop Publications

Preference-Based Distributed Welfare Maximization: A Game-Theoretic Approach
Antoine Bergerault⁺, Anna Maddux, Andreas Schlaginhausen, and Maryam Kamgarpour.
International Conference on Machine Learning (ICML) Workshop, 2026.

Eliciting Truthful Feedback for Preference-Based Learning via the VCG Mechanism

Leo Landolt*, Anna Maddux, Andreas Schlaginhaufen, Saurabh Vaishampayan, and Maryam Kamgarpour.

Artificial Intelligence and Statistics Conference (AISTATS), 2026.

No-Regret Learning in Stackelberg Games with an Application to Electric Ride-Hailing

Anna Maddux*, Marko Maljkovic*, Nikolas Geroliminis, and Maryam Kamgarpour.

IEEE Conference on Decision and Control (CDC), 2025.

Multi-Agent Learning in Contextual Games under Unknown Constraints

Anna Maddux and Maryam Kamgarpour.

Artificial Intelligence and Statistics Conference (AISTATS), 2024.

Reinforcement learning for scaffold-free construction of spanning structures

Gabriel Vallat*, Jingwen Wang, Anna Maddux, Maryam Kamgarpour, and Stefana Parascho.

ACM Symposium on Computational Fabrication (ACM SCF), 2023.

Data-driven behaviour estimation in parametric games

Anna Maddux, Nicolò Pagan, Giuseppe Belgioioso, and Florian Dörfler.

International Federation of Automatic Control (IFAC), 2023.

Working Papers

Learning Stackelberg Equilibria from Preferences over Unknown Multi-Objective Trade-offs

Marko Maljkovic, Anna Maddux, Nikolas Geroliminis, and Maryam Kamgarpour.

Seminars and Conference Presentations

IEEE Conference on Decision and Control Conference talk; Rio de Janeiro, Brazil.	Dec 2025
NCCR Phase II, ETH Zürich Poster presentation; Zürich, Switzerland.	Nov 2025
PhD seminar, Université de Lorraine Invited talk; Metz, France.	Nov 2024
Workshop on Learning in Games, Institut de Mathématiques de Toulouse Poster presentation; Toulouse, France.	Jul 2024
Artificial Intelligence and Statistics Conference (AISTATS) Poster presentation; Valencia, Spain.	May 2024
PhD seminar, EPFL Seminar talk; Lausanne, Switzerland.	Oct 2023

Teaching Experience and Supervision

University Courses

Teaching Assistant

Course	Institution	Terms
Foundations of Artificial Intelligence	EPFL	Fall 2022–2025
Multi-Agent Learning and Control	EPFL	Spring 2024
Analysis 2	ETH Zürich	Spring 2020
Microeconomics 1	University of Freiburg	Fall 2015, 2016
Microeconomics 2	University of Freiburg	Spring 2016

Student Supervision

All projects at EPFL, Lausanne.

Semih Zaman, Semester project

Competitive Multi-agent RL with Coupling Constraints.

Co-supervision with Philip Jordan and Maryam Kamgarpour.

Sep 2025 – Jan 2026

Leo Landolt , Master's thesis VCG mechanism for truthful demand response via preference feedback. <i>Co-supervision with Andreas Schlaginhaufen, Saurabh Vaishampayan, and Maryam Kamgarpour.</i>	Jan 2025 – Aug 2025
Hana Catic , Semester project Finite time convergence to an ϵ -efficient Nash equilibrium in potential games. <i>Co-supervision with Reda Ouhamma and Maryam Kamgarpour.</i>	Sep 2024 – Jan 2025
Antoine Bergerault , Semester project Learning to maximize the social welfare from preference feedback. <i>Co-supervision with Andreas Schlaginhaufen and Maryam Kamgarpour.</i>	Feb 2024 – Jun 2024
César Camus-Emschwiller , Semester project Learning in potential games with coupling constraints - Practical application. <i>Co-supervision with Gabriel Vallat and Maryam Kamgarpour.</i>	Feb 2024 – Jun 2024
Julien Drouet , Semester project Learning in Potential Games with Coupling Constraints. <i>Co-supervision with Gabriel Vallat and Maryam Kamgarpour.</i>	Feb 2024 – Jun 2024
Sabri El Amrani , Semester project Reinforcement Learning from Human Feedback for the Construction of Spanning Structures. <i>Co-supervision with Andreas Schlaginhaufen and Maryam Kamgarpour.</i>	Sep 2023 – Jan 2024
Dana Afazeli , E3 summer internship <i>Co-supervision with Kai Ren and Maryam Kamgarpour.</i>	Jun 2023 – Sep 2023
Titouan Renard , Master's thesis Provable Convergence Guarantees for Constrained Inverse Reinforcement Learning. <i>Co-supervision with Andreas Schlaginhaufen, Tingting Ni, and Maryam Kamgarpour.</i>	Feb 2023 – Jul 2023
Riccardo Ruzza , Bachelor project <i>Co-supervision with Giulio Salizzoni and Maryam Kamgarpour.</i>	Feb 2023 – Jun 2023
Julien Gouraud , Bachelor project <i>Co-supervision with Giulio Salizzoni and Maryam Kamgarpour.</i>	Feb 2023 – Jun 2023
Gabriel Vallat , Master's thesis Reinforcement learning for scaffold-free construction of spanning structures. <i>Co-supervision with Maryam Kamgarpour and Stefana Parascho.</i>	Sep 2022 – Jan 2023
Dorsa Majdi , E3 summer internship <i>Co-supervision with Maryam Kamgarpour.</i>	Jun 2022 – Sep 2022

Science Education and Outreach

Swiss TecDays: “Mathematics of the cities of the future”

Presentations for high-school students at the following schools in Switzerland:

Kantonsschule Hohe Promenade, Zürich	Nov 2025
Gymnasium Neufeld, Bern	Mar 2025
Kantonsschule Reussbühl, Reussbühl	Nov 2024
Kantonsschule Wohlen, Wohlen	Oct 2024
NCCR Automation: “A researcher in my class”	Feb 2024
Presentation at Kantonsschule Beromünster, Beromünster, Switzerland.	

Academic Service

Peer Review

International Conference on Learning Representations (ICLR), International Conference on Machine Learning (ICML) Workshop; Conference on Artificial Intelligence and Statistics (AISTATS); International Federation of Automatic Control (IFAC); European Control Conference (ECC); IEEE Conference on Decision and Control (CDC).

Conference and Workshop Organization

Workshop Organizer | National Centre of Competence in Research (NCCR) Automation junior retreat
Beatenberg, Switzerland. Led the Entrepreneurship and Innovation workshop, developing content and inviting speakers. Apr 2024

Fellowships and Awards

Accenture Female Talent Program

Mentoring program for women in STEM and economics.

Mar 2020 – Oct 2020

Fellow, Friedrich Naumann Foundation for Freedom

Academic scholarship (awarded to 1% of German university students).

Oct 2017 – May 2021

References

Maryam Kamgarpour

Associate Professor, Sycamore Lab, EPFL
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